

Analysis of Exoplanet Data:
Predicting Habitability Candidates using Machine Learning

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Abstract

With over 6 000 confirmed exoplanets discovered to date, the vast majority of planetary datasets remain incomplete, making the manual prioritization of habitability candidates resource-intensive. This project develops a ‘Two-Hurdle’ machine learning model to predict exoplanet habitability by utilizing stellar, system, and limited planetary data from the NASA Exoplanet Archive. Addressing the extreme class imbalance within the dataset, the model first employs an XGBoost classifier to filter planets within the Habitable Zone (HZ) around its host star. Then, Random Forest regressor is used to predict an Earth Similarity Index (ESI) score. The final model combines these outputs into a prioritized habitability ranking. Results demonstrate high performance against the ‘Gold Standard’ Habitable Worlds Catalog, achieving a Precision at 5 of 80% and a Precision at 10 of 60%. While testing on Earth’s own solar system’s data revealed inherent limitations in current theoretical habitability metrics, especially regarding Earth’s own unique profile, the model successfully filters out inhospitable candidates like Mars and Venus. This tool, served via a FastAPI and React.js dashboard, offers a method for astronomers to prioritize targets for further scientific study.

Introduction

Humans have barely scratched the surface of the vast expanse of what lies in the universe. Over the years, astrophysicists have developed various techniques and tools for detecting stars, planets, and more across the universe. As such, we have aggregated data on solar systems, host stars, and their exoplanets (Christiansen et al., 2025, n.p.). An exoplanet is a planet that does not belong to Earth’s solar system. Overall, we have discovered a lot more stars than planets due to their luminosity making them easier for our instruments to detect. Generally

speaking, it is through analyzing data from host stars that scientists discover planets. Through detecting anomalies in patterns of starlight caused by planets' transit between the sun and the instrument of detection, or finding evidence of planetary gravitational impact on its host star, we can discover new exoplanets (Christiansen et al., 2025, n.p.). Discovery, however, is only one piece of the puzzle. After discovery, it takes lots of time, resources, and effort to measure and catalog various metrics about planets such as their radius, mass, equilibrium temperature, orbital characteristics, and more. As a result, the dataset for many exoplanets remains incomplete (NASA Exoplanet Science Institute).

This is where my project comes in. I aim to utilize machine learning techniques to predict which already discovered exoplanets are high candidates for habitability using their system, stellar, and limited planetary data, thus allowing for easier prioritization of which exoplanets to investigate and study further. I implemented the backend in Python, using various data analysis and machine learning libraries. Specifically, I used Pandas (*User Guide*, n.d.), NumPy (*NumPy*, n.d.), and Scikit-Learn (*scikit-learn*, n.d.) for analysis and matplotlib pyplot (*matplotlib.pyplot*, n.d.) for visualization. Once my model was finalized, I used joblib (*Joblib: Running*, n.d.) to save it and FastAPI (*FastAPI*, n.d.) to serve my results to the frontend. To implement my frontend, I used React.js (*React*, n.d.) and css.

Literature Review

The NASA Exoplanet Archive is an online catalog of stellar and exoplanet data that is made available for public use to encourage data exploration. Prior to its creation, there were other datasets available such as the Extrasolar Planet Encyclopedia, and the Exoplanet Orbit Database, however, none of these were as comprehensive as the NASA Exoplanet Archive

(Akeson et al., 2013, pp. 2-3). The NASA Exoplanet Archive serves as a centralized, standardized datasource for an array of data analysis on various astronomical problems. A team of astronomers aggregate the data from peer-reviewed research papers on stars and exoplanets to populate the datasets (Akeson et al., 2013, p. 4). As interest in the field of exoplanets grew, the archive has amassed over 6 000 exoplanets, around 5 000 more confirmed exoplanets than it had when it first began. As such, the infrastructure evolved to handle the influx of data from missions like TESS and Gaia (Christiansen et al., 2025, n.p.). Christiansen et al. (2025, n.p.) highlight that the archive has been transitioning its infrastructure to make the data easily accessible for programmatic usage through a Table Access Protocol (TAP) API for analysis.

While the NASA Exoplanet Archive provides access to the data, further research was required to determine the standard metrics for what constitutes a habitability candidate. Two main metrics were found: the planet's orbit being within its host star's Habitable Zone (HZ) (Carney, 2025, n.p.) and a high Earth Similarity Index (ESI) score (Schulze-Makuch et al., 2011, n.p.). The HZ is the range of radii around a star in which temperatures are such that a planet orbiting within the zone could sustain liquid water. Kopparapu et al. (2013, n.p.) use water vapor and carbon dioxide absorption data to improve upon previous methods for calculating the inner and outer boundaries of the zone. Shown in Figure 1, they found that the optimal way of calculating the HZ is by first determining the incident stellar flux (S_{eff}) of the host star based on the effective temperature (T) and the constant representing our Sun's stellar flux ($S_{\text{eff}\odot}$). The coefficients (a , b , c , d) for the inner and outer radii were determined through their modeling of what best matches real-world data. The distance is then calculated by taking the square root of the quotient achieved by dividing the star's luminosity by our Sun's luminosity, and then by its S_{eff} (Kopparapu et al., 2013, n.p.).

Figure 1: Stellar flux and HZ distance calculations (Kopparapu et al., 2013, n.p.)

$$S_{\text{eff}} = S_{\text{eff}\odot} + aT_{\star} + bT_{\star}^2 + cT_{\star}^3 + dT_{\star}^4,$$

$$d = \left(\frac{L/L_{\odot}}{S_{\text{eff}}} \right)^{0.5} \text{ AU},$$

Proposed by Schulze-Makuch et al. (2011, n.p.), the Earth Similarity Index (ESI) was introduced as a multi-parameter method to characterize the physical similarity of exoplanets to Earth on a scale of 0 to 1. The calculation of the ESI is based on four fundamental physical properties of a given planet: radius, bulk density, escape velocity, and surface temperature. Shown in Figure 2, it is defined as the weighted geometric mean of the relative differences between an exoplanet and Earth. Where x_i is a planetary parameter, x_{i0} is the corresponding reference value for Earth, and w_i is the weight exponent determined experimentally by Schulze-Makuch et al. (2011, n.p.) for each parameter. They used a 0.8 ESI score as a threshold and solved for the weight exponents for each parameter, resulting in the values of 0.57 for radius, 1.07 for bulk density, 0.70 for escape velocity, and 5.58 for surface temperature. While ESI is a useful calculation for approximating physical similarity to Earth, it does not guarantee habitability.

Figure 2: ESI Calculation (Planetary Habitability Laboratory, 2024, n.p.)

$$ESI = \prod_{i=1}^n \left(1 - \left| \frac{x_i - x_{io}}{x_i + x_{io}} \right| \right)^{\frac{w_i}{n}}$$

To establish a benchmark for evaluating habitability predictions, the Planetary Habitability Laboratory (PHL) at the University of Puerto Rico, created and maintains the Habitable Worlds Catalog (HWC). The HWC is a standardized dataset of exoplanets generally ranked as ‘potentially habitable’. It uses various metrics including ESI scores, as well as expert analysis by a team of astronomers to classify exoplanets as being candidates for habitability (Planetary Habitability Laboratory, 2024). Although it serves as a ‘Gold Standard’ for this problem, it is still a theoretical approximation. The only planet we have concrete evidence of being habitable is Earth, and therefore there are no empirical ‘ground truth’ metrics that can be utilized for this problem.

Overall, habitability candidate classification using these standard metrics requires specific planetary and stellar data in order to calculate. Even still, these classifications are not always correct given the complexity of the problem, the lack of data, and the lack of a true ‘ground truth’. As such, using machine learning techniques to take the data that we have and predict which planets have a high likelihood of being habitability candidates in order to prioritize their further investigation is a worthwhile addition to the field.

Methodology

Preprocessing

The first step in my preprocessing pipeline was to initialize a virtual Python environment within which to install any packages I required and run my code. After getting that set up, I moved on to data ingestion and cleaning. The main source of data for this project was the ‘Planetary Systems Composite Parameters’ table from the NASA Exoplanet Archive (NASA Exoplanet Science Institute, 2020). I decided to use this table over the ‘Planetary Systems’ table as it is an aggregated version of the latter table, where there is only one row in the table per planet and thus more fields are filled-in per row with the expert-selected, most widely accepted data for the given planet. The latter table has one row per planet reference, thus leading to many entries for a given planet and potential inconsistencies across rows (NASA Exoplanet Science Institute, 2020). Using the aforementioned TAP API, I ingested the data into a Python Pandas dataframe (NASA Exoplanet Science Institute, 2025). By referencing the data column definitions on the NASA Exoplanet Archive website (NASA Exoplanet Science Institute, 2020) and analyzing the distributions and null counts of the columns, I removed columns that were irrelevant to the problem (e.g. columns linking to the source reference paper for each data attribute, columns describing discovery information) as well as columns with a majority null entries. After this, I was left with a much cleaner, more manageable dataset. Additionally, in order to prevent data leakage, I ensured that the features used to calculate my target variables (HZ and ESI score) were removed from my training feature set. This exploratory data analysis, however, revealed a major class imbalance within the dataset. Among the over 6 000 rows in the data, only around 70 were classified as ‘potentially habitable’ by the HWC (Planetary Habitability Laboratory, 2024). This defines a $\sim 1\%$ ratio of habitable to non-habitable planets in the dataset, as I did confirm that all the planets in the HWC are indeed in the dataframe.

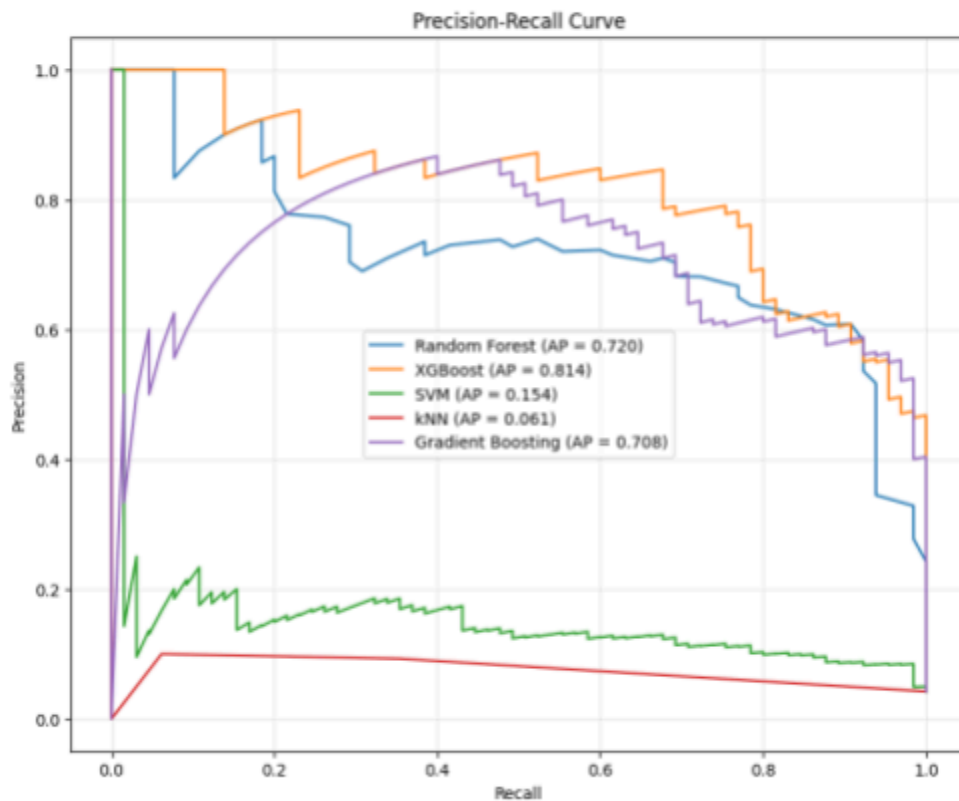
Initially, I attempted standard methods for tackling class imbalance. I tried artificially duplicating rows of the minority class, using log-transform to make the distribution more normalized, and applying the SMOTE algorithm to generate synthetic samples of the minority class to balance out the dataset (*SMOTE for Imbalanced*, 2026). However, none of these methods led to promising results. Due to the extreme imbalance, they all performed poorly and introduced far too much bias into the dataset to be viable options. Further research into the problem led me to the concept of a ‘Two-Hurdle Model’. Initially developed by John Cragg (1971, n.p.) for economic statistical models involving the demand for goods, the concept is applicable in a wide range of domains. When a dataset has a high ratio of ‘zero values’ to values of interest, standard regressors are unable to perform well. Cragg tested various methods of modeling using datasets with a high ratio of ‘zero values’, and found that by applying a classifier first to filter out some of the ‘zero values’, and then applying a regressor, it increases overall performance (Cragg, 1971, 842-844).

Classifier Testing

To implement the first hurdle of the model, I evaluated several classification algorithms to act as a binary filter, predicting whether an exoplanet resides within its host star’s HZ. The target variable was engineered using the aforementioned Kopparapu et al. (2013, n.p.) boundaries, which define the HZ based on stellar effective temperature and luminosity. I tested the following algorithms: Random Forest, XGBoost, Support Vector Machine (SVM), k-Nearest Neighbors (kNN), and Gradient Boosting. Due to the class imbalance causing a high accuracy for all models, I opted to use the Precision-Recall Curve to visualize the models’ performances instead of the ROC-AUC curve. The “large number of TNs (true negatives) from the majority class can dilute the impact of FPs (false positives)” (Sun, 2025), meaning that accuracy and

ROC-AUC can often provide a misleadingly optimistic view of a model's performance on highly imbalanced datasets such as this one. Figure 3 shows that XGBoost achieved a high recall without as big of a drop in precision, meaning that it performed best at identifying planets within their host star's HZ, without having as many false positives.

Figure 3: Performance of Classifiers



This conclusion is supported by comparing the models' overall F1 scores in Figure 4. It can be seen that XGBoost has the highest overall F1 score at 0.7634, indicating an overall balance in high performance between precision and recall. Additionally, the accuracy scores are high for all models, even kNN which performed extremely poorly due to the 'Curse of Dimensionality' and its poor ability to find neighbours in high-dimensional space. As mentioned, this is due to the high class imbalance and the fact that if a model ranks every point as being part of the majority class, it will still achieve high accuracy despite neglecting the minority class.

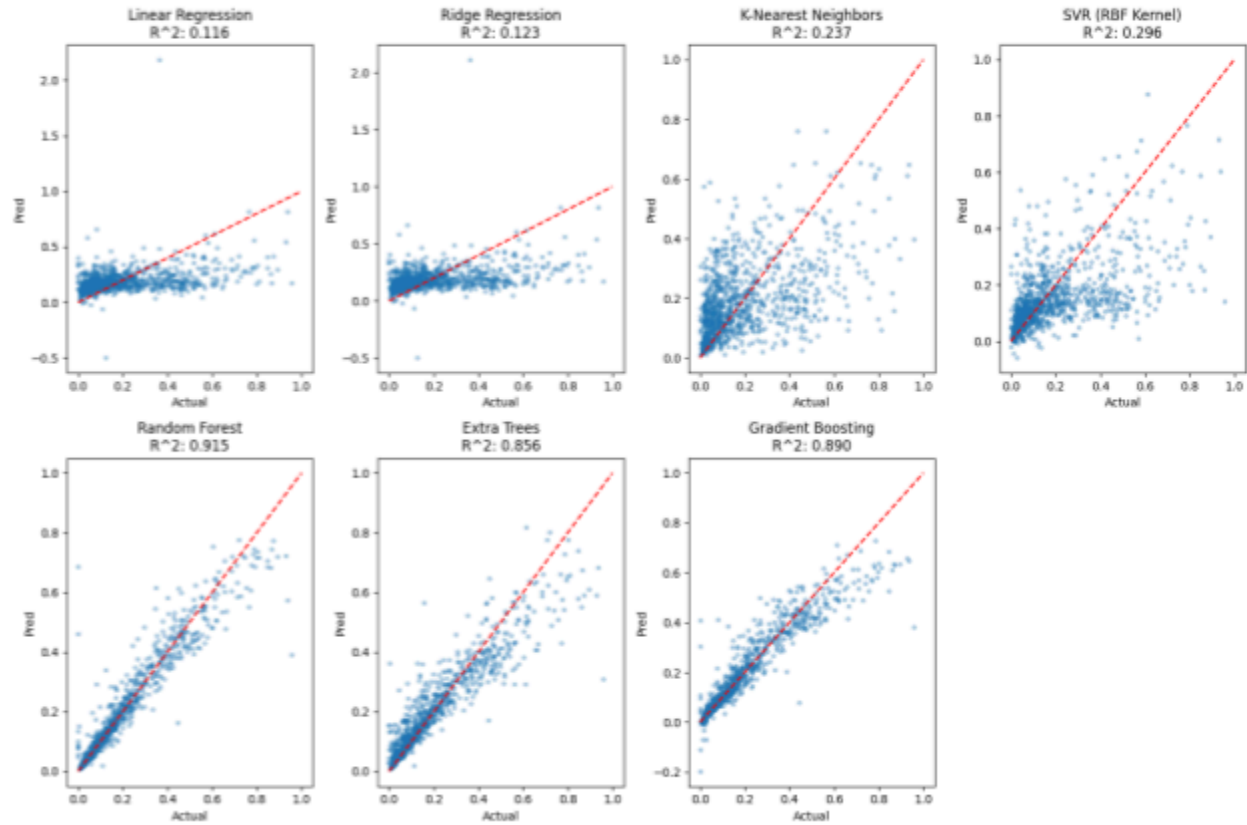
Figure 4: Classifier Evaluations

	Accuracy	Precision	Recall	F1 Score
Random Forest	0.964286	0.777778	0.215385	0.337349
XGBoost	0.979870	0.757576	0.769231	0.763359
SVM	0.791558	0.125731	0.661538	0.211302
kNN	0.951299	0.000000	0.000000	0.000000
Gradient Boosting	0.972727	0.848485	0.430769	0.571429

Overall, the XGBoost algorithm was shown to perform best at predicting whether a given exoplanet exists within its host star's HZ. As such, XGBoost was chosen to be used as the first step in the combined 'Two-Hurdle Model'.

Regressor Testing

For the model's second hurdle, I needed to find the best performing regression algorithm to predict ESI scores. Using the formula and parameter weights determined by Schulze-Makuch et al. (2011), I calculated my ESI target variable from the planet radius, density, escape velocity, and surface temperature in the dataset. Then, I tested a variety of regression models to determine which could most accurately predict these scores using the selected features. The algorithms tested were Linear Regression, Ridge Regression, kNN, Support Vector Regression (SVR), Random Forest, Extra Trees, and Gradient Boosting. To evaluate model performance, I used the R^2 score. Figure 5 shows each model's predicted value plotted against their actual value, as well as a line of the ideal scenario (where prediction matches actual perfectly). Visually, one can see that Random Forest, Extra Trees, and Gradient Boosting appear to perform the best with the plotted points lying nearest the line.

Figure 5: Regressor Performances Visualized

This analysis is supported by the models' R^2 scores, shown in Figure 6. Random Forest has the highest R^2 score at 0.9150, indicating a good fit of the model on the data.

Figure 6: Regressor Performance Summary

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--- Model Performance Summary (R^2 Score) ---
Random Forest      : 0.9150
Gradient Boosting  : 0.8899
Extra Trees        : 0.8559
SVR (RBF Kernel)   : 0.2956
K-Nearest Neighbors : 0.2373
Ridge Regression    : 0.1233
Linear Regression   : 0.1165

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Overall, the Random Forest Regressor performed best at predicting the ESI scores of exoplanets in the dataset.

Combined Model

Through these tests, I determined that the best method of forming my ‘Two-Hurdle Model’ would be to first narrow the dataset using an XGBoost classifier to predict which planets fall within the HZ of their host star, and then use the Random Forest regressor to predict ESI scores on the subset of the dataset. Subsequently, I multiplied the classifier’s probability of being in the HZ by the regressor’s predicted ESI score to obtain an overall potential habitability score through which to rank the exoplanets. In order to maintain consistency across the models, I implemented a single 80/20 train-test split and used stratified sampling on the HZ target to ensure that the training and testing sets both had a proportional representation of habitable candidates (*scikit-learn*, n.d.).

First, the XGBoost classifier was trained on the full scaled training set to implement the first hurdle filter. Second, the Random Forest regressor was trained specifically on the subset of planets that the HZ target identified as habitable. This allowed the regressor to learn the nuances of ‘Earth-like’ scores from a more balanced and relevant subset of data, rather than being skewed by the thousands of non-habitable ‘zero-value’ entries. Then, the Overall Habitability Potential scores were calculated, and the top 10 ranked planets in the training set were output, as shown in Figure 7. These planets all had extremely high HZ probabilities, as expected given the two-stage approach. They also had fairly comparable predicted and real ESI scores.

Figure 7: Results of ‘Two-Hurdle Model’ on Training Set

Top 10 Predicted Habitable Candidates:				
	Planet Name	Overall Habitability	Potential	\
1900	K2-18 b		0.845315	
4829	Wolf 1069 b		0.840111	
3849	Wolf 1061 c		0.814425	
2303	TRAPPIST-1 g		0.805246	
3932	L 98-59 f		0.794924	
5389	GJ 667 C e		0.783603	
2632	GJ 251 c		0.782460	
3589	Kepler-1545 b		0.781461	
850	Kepler-442 b		0.770337	
2334	TRAPPIST-1 h		0.759296	

	Habitable Zone Probability	Predicted ESI	Real ESI
1900	0.986720	0.856692	0.811327
4829	0.951702	0.882746	0.871302
3849	0.977746	0.832962	0.857972
2303	0.998361	0.806568	0.603901
3932	0.986850	0.805517	0.915748
5389	0.986202	0.794566	0.625915
2632	0.987416	0.792431	0.675713
3589	0.992758	0.787162	0.702933
850	0.951292	0.809780	0.861557
2334	0.907449	0.836736	0.458599

Results and Discussion

In order to evaluate the performance of my model overall, I employed a few techniques. One was to determine the Precision at various k values when comparing my rankings to the ‘Gold Standard’ HWC rankings. Another was to test the model using Earth, Venus, and Mars’ values to see how they would score, since our solar system’s planets are not included in the NASA Exoplanet Archive.

First, I determined that there were 18 planets from the HWC in the test set I was evaluating my model on. As such, these 18 planets served as the ‘ground truth’ for calculating the model's performance. By comparing my model's prioritized rankings against these known

candidates, I was able to calculate the Precision at K. Figure 8 shows that the model achieved a precision of 4 out of 5 for the top 5 spots, 6 out of 10 for the top 10, and 10 out of 20 for the top 20. This indicates that the machine learning pipeline is highly effective at identifying and ranking the most promising candidates similarly to the experts at the PHL.

Figure 8: Precision @ k for Overall Model Performance

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--- Model vs. PHL Consensus Evaluation ---
Precision@ 5: 80.00%
  Validated Matches: K2-18 b, TRAPPIST-1 g, Wolf 1061 c, Wolf 1069 b
Precision@10: 60.00%
  Validated Matches: GJ 667 C e, K2-18 b, Kepler-442 b, TRAPPIST-1 g, Wolf 1061 c, Wolf 1069 b
Precision@20: 50.00%
  Validated Matches: GJ 1002 c, GJ 667 C e, K2-18 b, K2-9 b, Kepler-1606 b, Kepler-442 b, Kepler-62 f, TRAPPIST-1 g, Wolf 1061 c, Wolf 1069 b

--- Ranked high by ML, but not in PHL ---
['L 98-59 f', 'GJ 251 c', 'Kepler-1545 b', 'TRAPPIST-1 h', 'TOI-1266 d']

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Using the NASA website, I gathered the required attributes for Earth (*Facts about Earth*, n.d.), Venus (*Venus Facts*, n.d.), Mars (*Mars Facts*, n.d.), and the Sun (*Our Sun: Facts*, n.d.) to use them to test my model. Many of the units used in astrophysics for planetary measurements are benchmarked against Earth, where Earth's value is defined as 1 unit and other planets are compared as a proportion more or less than (or equal to) Earth. The same can be said for stellar measurements being benchmarked against the Sun (NASA Exoplanet Science Institute, 2020). As such, many of the values were simply 1. Figure 9 shows the results of this test. The model did not perform as well as expected on Earth's data, however, it did correctly identify that Venus and Mars are not good candidates for habitability. One reason why it may not have performed as well as expected on Earth is that these metrics for calculating habitability candidates are not perfect. They are approximations in themselves, and are purely theoretical. We have not found definitive proof of habitability on any planet besides Earth, and since the NASA Exoplanet Archive is specifically meant to house exoplanet data, these approximations are a kind of 'best guess' for 'second place'. The model is learning patterns based on the data it has available, but due to the

uniqueness of Earth, we haven't found enough planets that are similar enough to Earth which has caused this low score.

Figure 9: Model Performance on Earth, Venus, and Mars Data

	Planet	Overall_Habitability_Score
0	Earth	0.606203
1	Venus	0.197464
2	Mars	0.394896

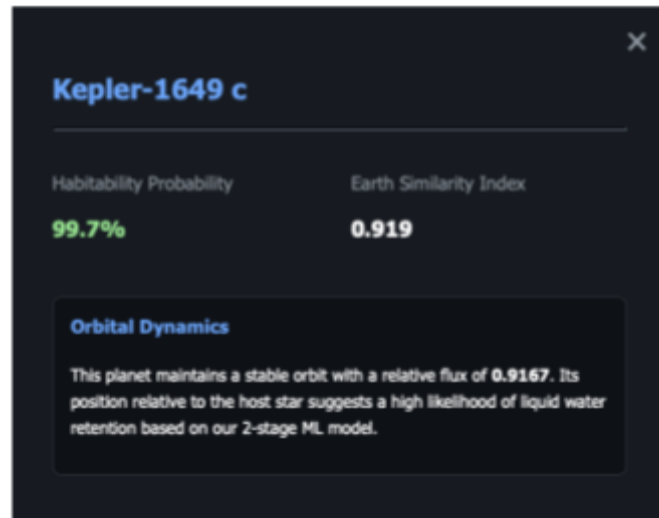
Upon completing my combined model, I used Joblib (*Joblib: Running*, n.d.) to save my model and FastAPI (*FastAPI*, n.d.) to run my backend and serve my model results to my frontend. I implemented a simple dashboard UI using React.js and CSS to display my model's overall results, as shown in Figure 10.

Figure 10: React.js Frontend Model Visualization

NASA Exoplanet Archive Data Analysis				
Top Candidates Data Insights				
CANDIDATE NAME	HABITABILITY POTENTIAL PREDICTION	HABITABILITY ZONE PREDICTION	ESI PREDICTION	PHL COMPARISON
Kepler-1649 c	91.7%	100%	0.919	✓ CONFIRMED
TOI-700 d	90.1%	100%	0.901	✓ CONFIRMED
TRAPPIST-1 d	89.6%	100%	0.900	✓ CONFIRMED
TOI-700 e	89.3%	100%	0.894	✓ CONFIRMED
Teegarden's Star b	89.1%	99%	0.900	✓ CONFIRMED
Kepler-1512 b	89.0%	99%	0.896	✗ REFUTED
Kepler-296 e	86.2%	100%	0.865	✓ CONFIRMED

Figure 11 shows the further insights depicted when the user selects a row in the table.

Figure 11: Planet Information Pop-Up



Conclusion

Overall, my model performed well when compared to the ‘Gold Standard’ HWC dataset, with high Precision at k values on the test set. However, when Earth itself was used as a test point, the model only ranked it a 0.606. This showcases the broad limitations in the field of predicting habitability, due to the fact that we do not have any concrete evidence for ‘ground truth’ habitable planets besides Earth. Due to this, our ‘Gold Standard’ is still an approximation based on our current mathematical and scientific capabilities. The model does, however, achieve its goal of predicting habitability candidates for further investigation using limited planetary, stellar, and system data.

Next steps for the project would be to further develop the frontend user interface. I plan to implement visualization of exoplanets, as well as incorporate more data visualization in the form of graphs and charts so that users can get a clearer picture of the data behind the model. Additionally, I will implement a feature where users can input new planet data and run the model to get its prediction for the new planet’s habitability score. Additionally, when more atmospheric

data is available for exoplanets in the NASA Exoplanet Archive, I would like to incorporate that into the model and see how it alters the results.

Throughout this project, I learned how tackle the problem of class imbalance in machine learning models, how to successfully analyze data in an unfamiliar domain by researching the field, how to build, test, and evaluate a machine learning model, and how to successfully connect a backend and frontend to achieve a well-rounded final project.

References

- Akeson, R. L., Chen, X., Ciardi, D., Crane, M., Good, J., Harbut, M., Jackson, E., Kane, S. R., Laity, A. C., Leifer, S., Lynn, M., McElroy, D. L., Papin, M., Plavchan, P., Ramírez, S. V., Rey, R., von Braun, K., Wittman, M., Abajian, M., . . . Zhang, A. (2013). The NASA exoplanet archive: Data and tools for exoplanet research. *Publications of the Astronomical Society of the Pacific*, 125(930), 989-999. <https://doi.org/10.1086/672273>
- Carney, S. (Ed.). (2025, September 3). *The Habitable Zone*. NASA. Retrieved April 29, 2026, from <https://science.nasa.gov/exoplanets/habitable-zone/>
- Christiansen, J. L., McElroy, D. L., Harbut, M., Ciardi, D. R., Crane, M., Good, J., Hardegree-Ullman, K. K., Kesseli, A. Y., Lund, M. B., Lynn, M., Muthiar, A., Nilsson, R., Oluyide, T., Papin, M., Rivera, A., Swain, M., Susemihl, N. D., Tam, R., van Eyken, J., & Beichman, C. (2025). The NASA exoplanet archive and exoplanet follow-up observing program: Data, tools, and usage. *The Planetary Science Journal*, 6(8), 186. <https://doi.org/10.3847/psj/ade3c2>
- Cragg, J. G. (1971). Some statistical models for limited dependent variables with application to the demand for durable goods. *Econometrica*, 39(5), 829-844. <https://doi.org/10.2307/1909582>
- Facts about Earth*. (n.d.). NASA. Retrieved April 29, 2026, from <https://science.nasa.gov/earth/facts/>
- FastAPI*. (n.d.). FastAPI. Retrieved April 29, 2026, from <https://fastapi.tiangolo.com>
- Joblib: Running Python functions as pipeline jobs*. (n.d.). Joblib. Retrieved April 29, 2026, from <https://joblib.readthedocs.io/en/stable/>

- Kopparapu, R. K., Ramirez, R., Kasting, J. F., Eymet, V., Robinson, T. D., Mahadevan, S., Terrien, R. C., Domagal-Goldman, S., Meadows, V., & Deshpande, R. (2013). HABITABLE ZONES AROUND main-sequence stars: NEW ESTIMATES. *The Astrophysical Journal*, 765(2), 131. <https://doi.org/10.1088/0004-637x/765/2/131>
- Mars Facts*. (n.d.). NASA. Retrieved April 29, 2026, from <https://science.nasa.gov/mars/facts/>
- matplotlib.pyplot*. (n.d.). Matplotlib. Retrieved April 29, 2026, from https://matplotlib.org/stable/api/pyplot_summary.html
- NASA Exoplanet Science Institute. (2020). Planetary Systems Composite Table. *NASA Exoplanet Archive*. Retrieved April 28, 2026, from <https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbIs&config=PS>
- NASA Exoplanet Science Institute. (2025, June 3). *Retrieving Exoplanet Archive Data With Table Access Protocol*. NASA Exoplanet Archive. Retrieved April 28, 2026, from <https://exoplanetarchive.ipac.caltech.edu/docs/TAP/usingTAP.html>
- NumPy reference*. (2024, August 19). NumPy. Retrieved April 29, 2026, from <https://numpy.org/doc/2.1/reference/index.html>
- Our Sun: Facts*. (n.d.). NASA. Retrieved April 30, 2026, from <https://science.nasa.gov/sun/facts/>
- Planetary Habitability Laboratory. (2024, March 21). *Habitable Worlds Catalog*. Planetary Habitability Lab at University of Puerto Rico. Retrieved April 29, 2026, from <https://phl.upr.edu/hwc>
- React*. (n.d.). React. Retrieved April 29, 2026, from <https://react.dev>
- Schulze-Makuch, D., Méndez, A., Fairén, A. G., von Paris, P., Turse, C., Boyer, G., Davila, A. F., António, M. R. de S., Catling, D., & Irwin, L. N. (2011). A two-tiered approach to

assessing the habitability of exoplanets. *Astrobiology*, 11(10), 1041-1052.

<https://doi.org/10.1089/ast.2010.0592>

scikit-learn. (n.d.). scikit-learn. Retrieved April 29, 2026, from <https://scikit-learn.org/stable/>

SMOTE for Imbalanced Classification with Python. (2026, February 10). Geeks for Geeks.

Retrieved April 29, 2026, from

<https://www.geeksforgeeks.org/machine-learning/smote-for-imbalanced-classification-with-python/>

Sun, P. (2025, May 22). *Stop using ROC Blindly in Classification*. Medium. Retrieved April 29, 2026, from

<https://medium.com/@pacosun/stop-using-roc-blindly-in-classification-4d754923f7a2>

User Guide. (n.d.). Pandas. Retrieved April 29, 2026, from

https://pandas.pydata.org/docs/user_guide/index.html

Venus Facts. (n.d.). NASA. Retrieved April 29, 2026, from

<https://science.nasa.gov/venus/venus-facts/>